

Introduction to Finding the Relative Extrema of Multivariable Functions

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I Introduction

Let $f(x, y)$ be continuous in a closed region. Then, it attains a absolute maximum or minimum in said region. We say that if f has either a maximum or minimum then it has an extrema. Similar to Calculus I, a relative extrema is an absolute extrema in a small region, and occurs at a critical point. A critical point is where $f_x = f_y = 0$ or when one or both of the derivatives are not defined.

II The Second Derivative Test (Multivariable Version)

To determine if $f(x, y)$ has a relative extrema or not, we can apply the second derivative test. If $f(x, y)$ twice differentiable at critical point (x_0, y_0) . Then, we can define

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2 \quad (1)$$

- If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then $f(x_0, y_0)$ is a relative minimum
- If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then $f(x_0, y_0)$ is a relative maximum
- If $D < 0$ and $f_{xx}(x_0, y_0)$, is NOT a relative extrema, then (x_0, y_0) is a called a saddle point.
- If $D = 0$, then this test is inconclusive.

III Example

Let $f(x, y) = x^3 - 9xy + y^3$. Then,

$$f_x = 3x^2 - 9y, f_{xx} = 6x, f_y = 3y^2 - 9x, f_{yy} = 6y, f_{xy} = -9$$

[†]Definitions and example problems from MATH 22005: Analytic Geometry and Calculus III instructed by Dr. Gang Yu. Kent State University.

First we set $f_x = f_y = 0$ and solve a system of equations. Our goal is to reduce it down to one variable and solve for the critical points. So,

$$\begin{cases} 3x^2 - 9y \\ 3y^2 - 9x \end{cases} \implies \begin{cases} y = \frac{1}{3}x^2 \\ y^2 = 3x \end{cases} \implies \left(\frac{1}{3}x^2\right)^2 = 3x \implies x(x^3 - 27) = 0$$

So $x = 0, 3$. At $x = 0, y = \frac{1}{3}(0)^2 = 0$ and at $x = 3, y = 3$ So, our two critical points are $(0, 0)$ and $(3, 3)$. Now we plug into our D function

$$D(0, 0) = (6(0)) \cdot (6(0)) - (-9)^2 = -81 < 0 \implies \text{saddle point}$$

$$D(3, 3) = (6(3)) \cdot (6(3)) - (-9)^2 = 243 > 0 \text{ and } f_{xx} = 18 > 0 \implies \text{relative minimum}$$

Now we simply plug in $(3, 3)$ into f to find the minimum value. Note that we do NOT plug in $(0, 0)$ because it is a saddle point.

$$f(0, 0) = (3)^3 - 9(3)(3) + (3)^3 = -27$$